**COMP 8130**
**Topics in Software Engineering**
**Winter 2026**

SCHOOL OF COMPUTER SCIENCE

# Assignment-2

**This is an individual assignment. Cheating and plagiarism in any part of the assignment will result in zero for the whole assignment**

In this second assignment, you need to develop a classifier to predict the intention in the code review questions. Select one multi-class prediction model from sklearn or any other machine learning libraries.

**Deadline: April 05, 2026 (11.59 PM)**
**Total Marks:** 10

**Reference Paper: Predicting Communicative Intention in Code Review Questions**

Dataset: icsme-questions-labeled.xlsx (available on the course website)

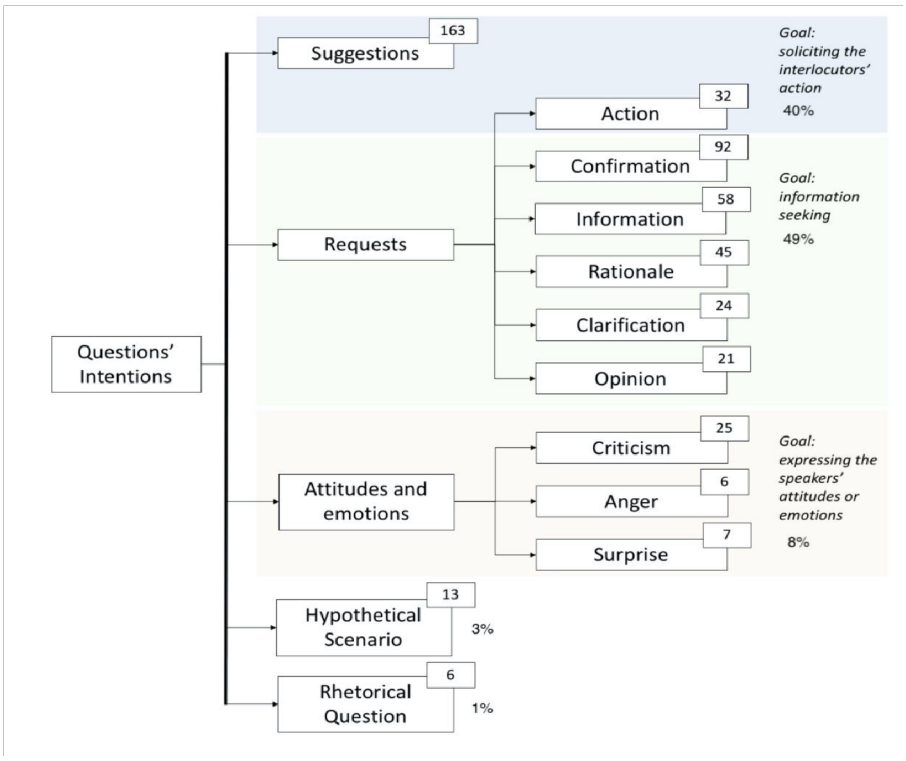
**Note:** Include at least one new feature other than only considering the words appearing

in those questions (you can consider more features!).

**Input:** A specific question in the code review comments

**Labels:** Consider the top-level categories

1. Suggestions
2. Requests
3. Attitudes and emotions
4. Hypothetical scenario
5. Rhetorical questions




**Examples from the provided dataset**

| inline-comment-id | # Comment | Question | Final Label |
|---|---|---|---|
| 84326dd1_566c7146 | 1 | is this what they intended? don't they really want $(TARGET_OUT_DATA_NATIVE_TEST)? | request for confirmation |
| 84326dd1_566c7146 | 2 | is this what they intended? don't they really want $(TARGET_OUT_DATA_NATIVE_TEST)? | surprise |
| 99d1f8e4_92b31cea | 3 | Don't we need to increment 'i' in the else case here to avoid an infinite loop? | suggestion |
| 193d089f_f5fac752 | 4 | i can't see anywhere where this is set to false. should we just adjust the single reference in bionic and then remove this? | suggestion |

**Tasks**

1. Generate a model with words as the feature, with text preprocessing step
2. Generate a model with words as the feature + your feature, without text preprocessing step
3. Generate a model with words as the feature + your feature, with text preprocessing step

4. Generate a model with words as the feature + your feature, with resampling and text preprocessing step

**Submission instructions**

Submit the code as a Jupyter notebook, data file(2) to train and test the technique, a document describing the experiment process (same format that you used for submitting project proposal) and the results of four models (Not more than four pages). **The document must contain an introduction describing the problem description, dataset description, experiment process, study results and references.**